

Laboratory Test Report — Core Tests, UPV Survey and Petrographic Analysis, Pier P2 Pier Cap

Nandira River Bridge Project — Package NRB-4

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From:	Suvarna Materials Testing Laboratory
To:	Meridian Engineering Consultants
CC:	Sagara Constructions Pvt. Ltd.

Subject: Core Tests, UPV Survey and Petrographic Analysis — Pier P2 Pier Cap

Testing carried out 18-Mar-2024 to 26-Mar-2024, per the scope agreed in Joint Site Inspection Record SIR-NRB4-001 and following the Crack Mapping Report CMR-NRB4-001.

1. CORE TESTS (COMPRESSIVE STRENGTH):

Three cores extracted from the central (highest crack density) zone of the pier cap, length-to-diameter ratio 2.0 (no correction factor required). Specified characteristic strength: 40 N/mm2 at 28 days (Grade M40, per Employer's Requirements Section 5).

Core	Raw Strength (N/mm2)	Corrected Strength (N/mm2)
C-1	44.6	44.6
C-2	43.1	43.1
C-3	45.0	45.0
Average corrected core strength:		44.2 N/mm2

ASSESSMENT: All three results exceed the specified characteristic strength; the concrete's bulk compressive strength is not implicated in the cracking.

2. ULTRASONIC PULSE VELOCITY (UPV) SURVEY:

Zone	UPV Range (km/s)	Average (km/s)	Classification
Central (cracked)	4.2 – 4.5	4.3	Good to Excellent
Edge (uncracked)	4.2 – 4.6	4.4	Good to Excellent

ASSESSMENT: UPV readings in the cracked central zone are consistent with the uncracked edge zone, indicating dense, homogeneous concrete below the surface, with no significant internal voiding or discontinuity attributable to the cracking.

3. PETROGRAPHIC EXAMINATION:

Thin sections prepared from Core C-1 (through a representative surface crack) examined under polarising microscope.

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- Crack depth measured at 12mm to 19mm from the top surface across the three sections examined; no crack observed to extend below approximately 20mm depth in any section.
 - No evidence of alkali-silica reaction (ASR) gel, sulfate attack products, or delayed ettringite formation at or adjacent to any crack examined.
 - Aggregate distribution, paste microstructure, and air content of the bulk concrete below the cracked zone are consistent with a properly proportioned and compacted Grade M40 mix; no evidence of segregation, excess water content, or abnormal aggregate reaction.
 - Carbonation depth measured at 2mm to 3mm, consistent with normal carbonation for concrete of this age and exposure; no evidence of reinforcement corrosion or corrosion-induced cracking (cracks do not align with, or extend to, the cover to reinforcement, measured at 45mm to 50mm at the locations examined, in accordance with the Specification).

4. OVERALL CONCLUSION:

The cracking is shallow (confined to approximately the top 20mm of the pier cap), is not associated with any deficiency in the bulk concrete's strength, density, or composition, and is not associated with alkali-silica reaction, sulfate attack, or reinforcement corrosion. These findings are consistent with surface (plastic or early-age drying shrinkage) cracking, most commonly associated with insufficient curing of the exposed surface in the days immediately following placement, rather than with a deficiency in the concrete mix, the in-place placement or compaction of the concrete, or the structural design. We were not instructed to, and have not, reviewed curing records for this pour, which fall outside the scope of this laboratory testing.

Issued by: Suvarna Materials Testing Laboratory